

Self Distillation using only supervised fine tuning: a duplex cascade case study

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Abstract—Fine-tuning a full-duplex speech model to emit conversational turn-taking tags (the *duplex cascade* micro-turn protocol) with plain supervised fine-tuning (SFT) regresses the model’s general capabilities: function calling, instruction following and world knowledge degrade because the off-policy training signal pushes the whole next-token distribution toward tag-heavy dialogue. We show that this regression can be avoided *without reinforcement learning and without on-policy rollouts* by borrowing the self-distillation idea of *Self-Distillation Fine-Tuning* (SDFT): the frozen original model acts as its own teacher, and the fine-tune optimizes a two-term loss—the weighted cross-entropy on the duplex tags exactly as before, plus a per-token forward-KL to the frozen base model on every other (content) position. The content term *anchors* the student to the pretrained distribution everywhere the tags are not explicitly supervised, which is precisely the mechanism that prevents regression. We train a QLoRA adapter on 10k teacher-derived duplex dialogues and evaluate against both the base model and the plain-SFT baseline. The distill model learns the protocol (96.4% tag accuracy) while keeping function calling and general instruction-following at base level (100% vs the SFT baseline’s 55.6% and 41.7%), and it stays within 1–2 pp of base on MMLU, HellaSwag, ARC and TruthfulQA, and above base on IFEval—where the SFT baseline drops broadly. The only measured regression is on multi-step math reasoning (GSM8K). These results establish per-token KL-anchoring to the frozen base as a practical, RL-free anti-forgetting mechanism for task-specific fine-tuning.

Index Terms—self-distillation, continual learning, supervised fine-tuning, full-duplex dialogue, turn-taking, catastrophic forgetting, function calling

I. INTRODUCTION

Large language models that power interactive systems are often post-trained on narrowly defined behavioral protocols. A representative example is the *duplex cascade* paradigm [2]: a conversational LLM is fine-tuned to emit special turn-taking tokens (“<—user is speaking—i”, “<—user finish speaking—i”, “<—user interruption—i”, etc.) that drive a VAD-free, full-duplex speech pipeline. Learning the protocol is easy—the supervision is cheap, synthetic and deterministic—but the fine-tune is destructive. In our pt-BR implementation of the duplex cascade, the plain-SFT model that learns the tags with 98% accuracy simultaneously loses most of its ability to call functions (100% → 55.6%) and follow general instructions (100% → 41.7%). This is catastrophic forgetting in its most practical form: the model becomes a good turn-taker and a bad assistant.

The conventional remedy is to make the post-training signal *on-policy*, via reinforcement learning (RL) with an explicit reward [7], [8]. But rewards are often unavailable, and on-policy RL (PPO/GRPO) is expensive and brittle. A recent

line of work, *Self-Distillation Fine-Tuning* (SDFT) [1], obtains the benefits of on-policy learning from demonstrations alone: the model is used in two roles—a *teacher* conditioned on a demonstration, and a *student* conditioned only on the query—and the training minimizes a KL divergence between the two on the student’s own trajectories. SDFT substantially reduces catastrophic forgetting compared to SFT, but it still requires on-policy rollouts (roughly 4× the wall-clock of SFT).

We show that for the duplex-cascade setting there is a much simpler instantiation that needs *neither RL nor rollouts*. Our key observation is that the “new skill” (the turn-taking protocol) and the “everything else” (the general capabilities we want to preserve) are *separable by token position*: the tags live at supervised positions, the content lives at the other positions. We therefore split the objective in two:

- **Tags:** the weighted cross-entropy to the supervised tag tokens, exactly as in the plain SFT recipe.
- **Everything else:** a per-token forward-KL to the *frozen base model* on content positions, so the student’s distribution is anchored to the pretrained policy wherever there is no explicit tag supervision.

The teacher is the frozen original model itself (conditioned on the same duplex sequence via the tag tokens in its input), so no second model and no rollouts are needed. Because the student is *base + LoRA*, the teacher forward is obtained with zero extra VRAM by temporarily disabling the adapters. We precompute the teacher’s top-*k* next-token logits once (offline cache), which removes the second forward pass entirely and makes the run no more expensive than SFT.

We evaluate on a pt-BR duplex-cascade model (Qwen3-4B-Instruct-2507, QLoRA, 10k teacher-derived dialogues). The resulting *distill* model matches the plain-SFT model on the protocol (96.4% vs 98% tag accuracy) while retaining base-level function calling, general instruction following, MMLU, HellaSwag, ARC and TruthfulQA, and *improving* on IFEval. The plain-SFT baseline regresses on all of them. The one caveat is GSM8K, where the distill model drops 11 pp below base—a limitation we discuss in Section VI.

II. BACKGROUND AND RELATED WORK

A. Catastrophic forgetting and off-policy fine-tuning

Sequential SFT is inherently off-policy: the model learns to imitate expert actions on a fixed offline distribution, and prior work has shown this leads to severe forgetting when the model is adapted to a new task or domain [4]–[6].

On-policy methods—training on trajectories induced by the model itself—forget substantially less [3], [7]–[9]. DAgger [3] established that on-policy imitation avoids the compounding-error failure mode of behavior cloning. More recently, “SFT memorizes, RL generalizes” [7] and related analyses [8] showed empirically that RL-based post-training retains general capabilities far better than SFT.

B. Self-distillation

Our method is most closely related to *Self-Distillation Fine-Tuning* (SDFT) [1]. SDFT builds the teacher by conditioning the model on an expert demonstration, $\pi(\cdot|x, c)$, and trains the student $\pi_\theta(\cdot|x)$ to match it on the student’s own rollouts:

$$\begin{aligned} L(\theta) &= D_{KL}(\pi_\theta(\cdot|x) \parallel \pi(\cdot|x, c)) \\ &= \mathbb{E}_{y \sim \pi_\theta} \sum_t \log \frac{\pi_\theta(y_t|y_{<t}, x)}{\pi(y_t|y_{<t}, x, c)}. \end{aligned} \quad (1)$$

The paper shows that on Tool Use, Science Q&A and Medical skill-learning, SDFT achieves higher new-task accuracy than SFT while keeping previous-capabilities performance near the base model, and that an EMA of the student is the most stable teacher. Our work differs in two ways. First, we do *not* run on-policy rollouts; instead the tags are supplied by a deterministic construction rule, so the only on-policy-sensitive component is the content, which we anchor to the frozen base distribution directly. Second, our teacher is the frozen base model (not a demonstration-conditioned variant), which is exactly the “keep everything else the same” reference the anti-forgetting objective requires.

Context distillation [11], [12] also uses a model conditioned on extra context as a teacher for the model without it, but typically distills offline from static contexts and does not combine a separate supervised tag term with a content-anchoring KL. *Learning by distilling context* [12] distills a prompt-conditioned teacher into the base model; our tag term is the analogous “hard” supervision and our content term the distillation.

C. Parameter-efficient fine-tuning

We train with QLoRA [14] (4-bit NF4 base, LoRA [13] rank 16, 16 α , dropout 0) via the Unsloth [15] training library, keeping the embedding and LM head trainable so the new special-token rows can be learned.

III. METHOD: DUPLEXCASCADE-DISTILL

A. Setup

Let the base model be π_0 (frozen) and the student π_θ be the same weights plus a LoRA adapter (θ = adapter parameters). The training corpus is a set of *duplex sequences* x produced by the duplex micro-turn builder: a dialogue is split into micro-turns and conversational special tokens ($\mathcal{T}$) are inserted deterministically at turn boundaries (e.g. “<—user finish speaking— ζ ” before every system turn, probabilistic interruptions/backchannels). Every position t of x is thus either a *tag* position (label $y_t \in \mathcal{T}$) or a *content* position (label y_t is ordinary text). User-side and header positions are masked (label $-\infty$), as in the original recipe.

B. Teacher content

The tags themselves come from the deterministic builder. The *content* comes from the teacher: the frozen base model, prompted with a *duplex-cascade trigger* system message (“you are a DuplexCascade full-duplex voice assistant... speak in short, natural micro-turns... no tags”), generates each assistant turn conditioned on the history. At first the base model does not follow the protocol faithfully—that is expected and irrelevant; we keep only its *content* and let the deterministic builder place the tags. This yields the aligned demonstration sequence x .

C. Loss

Let T be the set of tag positions and C the set of content positions. The student loss is

$$\begin{aligned} L(\theta) &= \underbrace{\sum_{t \in T} w_t \text{CE}(\pi_\theta(\cdot|x_{<t}), y_t)}_{\text{tags: as in plain SFT}} \\ &+ \underbrace{\lambda \sum_{t \in C} D_{KL}(\pi_0(\cdot|x_{<t}) \parallel \pi_\theta(\cdot|x_{<t}))}_{\text{content: anchor to the frozen base}}, \end{aligned} \quad (2)$$

where w_t are the per-special-token weights of the original recipe (e.g. “<—user finish speaking— ζ ” = 10, “<—user interruption— ζ ” = 5). The KL is in the forward direction and computed in cross-entropy form $-\sum_k p_0(k) \log p_\theta(k)$ over the teacher’s top- k ($k=32$) next-token support, with the student logits gathered at those ids. Because the teacher is conditioned on the same x (its input already contains the tags), its content distribution is the natural “what would the original model say in a duplex dialogue” reference. The single hyper-parameter λ trades tag fidelity against anchoring; we use $\lambda=1.0$.

D. Practical implementation

- **Teacher forward is free.** The student is base + LoRA, so the teacher is the same weights without adapters: one additional `no_grad` forward under `disable_adapter()` costs zero extra VRAM.
- **Offline teacher cache.** To avoid the second forward in the training loop (and the GPU-allocator pressure it creates at the 16 GB ceiling), we precompute the teacher’s top- k ids/logits for every position of every sequence once (one bf16 forward pass over the corpus, ≈ 36 min). Training then reads cached targets and runs at the plain-SFT cost. This is the difference between our ~ 10 s/step and the two-forward path.
- **Anchor set (D2 variant).** A small set of general-capability prompts (function calling, QA) is distilled the same way, pinning the LoRA delta to ≈ 0 outside the duplex protocol.
- **Artifact masking.** Following the SDFT “learned artifacts” fix [1], the first few content tokens after each “<—user finish speaking— ζ ” tag are left unsupervised so the student does not copy teacher surface artifacts.

IV. EXPERIMENTAL SETUP

A. Models

Base: Qwen/Qwen3-4B-Instruct-2507 (dense Qwen3, 151.9k vocab). Student: QLoRA (r= 16, $\alpha=16$, dropout 0, all linear modules, trainable embeddings and LM head), 7 duplex special tokens appended and Gaussian-initialized. Teacher: the same base, frozen. All models evaluated as bf16.

B. Data

10,000 pt-BR dialogues from the existing corpus had every assistant turn regenerated by the teacher (trigger prompt, temperature 0.7, top-p 0.95), then passed through the deterministic micro-turn builder (user chunks 1–7 tokens, system chunks 10–48, probabilistic pauses/interruptions/backchannels), yielding 10,249 training sequences at ≤ 1024 tokens. 300 anchor prompts (150 function calling, 150 general QA).

C. Training

2,000 optimizer steps, batch $1 \times \text{grad-accum } 16$ (effective 16), AdamW lr 10^{-4} (embeddings 2×10^{-4}), 20-step warmup, bf16, gradient checkpointing, $\lambda=1.0$ (chosen by a short sweep on $\lambda \in \{0.1, 0.3, 1.0\}$ using tag accuracy and KL drift). The sft baseline is the existing DuplexCascade-PT QLoRA fine-tune (same data family, no KL term).

D. Evaluation

Protocol: tag accuracy (argmax = supervised tag) on a held-out duplex eval split. *Capabilities*: 18 pt-BR function-calling prompts (native Qwen tool calls, greedy) and 12 general instruction/QA prompts. *Distribution drift*: mean per-token forward-KL (student || frozen base) on the content positions of the held-out split—a direct measure of “everything else stayed the same”. *Standard benchmarks*: HellaSwag, ARC-Challenge, MMLU (2000-example subsets), GSM8K (5-shot, 300), TruthfulQA (mc2), IFEval (40) via lm-evaluation-harness, raw prompts, no chat template, identical configs for all three models.

V. RESULTS

A. Protocol vs. capability retention

Table I shows the central result. The *distill* model learns the protocol (96.4% tag accuracy, only 1.6 pp behind the SFT baseline) while keeping function calling and general instruction-following at *base level* (100%), and a KL drift only slightly above the base floor. The plain-SFT baseline, by contrast, learns the protocol (98%) but regresses catastrophically on the format-following capabilities (function calling 55.6%, general 41.7%) and drifts $\sim 3.5\times$ further from the base distribution.

TABLE I
PROTOCOL AND CAPABILITY RETENTION (OUR PT-BR EVAL SETS).

	base	distill	sft
Tag accuracy (%)	0.0	96.4	98.0
Function calling (%)	100	100	55.6
General instruction/QA (%)	100	100	41.7
KL drift vs base (nats)	0.51	1.07	4.76

TABLE II
STANDARD BENCHMARKS (LM-EVALUATION-HARNESS 0.4.12; SUBSETS AS NOTED).

Benchmark	base	distill	sft
HellaSwag acc-norm ($n=2000$)	0.584	0.566	0.593
ARC-Challenge acc-norm ($n=2000$)	0.584	0.563	0.574
MMLU ($n=2000$)	0.705	0.698	0.660
GSM8K exact ($n=300$, 5-shot)	0.890	0.777	0.827
TruthfulQA mc2	0.626	0.649	0.601
IFEval loose / strict ($n=40$)	0.730/0.651	0.762/0.683	0.603/0.524

B. Standard benchmarks

Table II reports the lm-evaluation-harness results. The distill model is within $\sim 1\text{--}2$ pp of base on knowledge (MMLU), commonsense (HellaSwag) and reasoning (ARC), *above* base on factuality (TruthfulQA +2.3 pp) and instruction following (IFEval loose +3.2 pp), and at base on the format-following evals of Table I. The SFT baseline drops on MMLU (-4.5 pp), TruthfulQA (-2.5 pp), IFEval (-12.7 pp) and the custom sets. The one regression of the distill model is GSM8K (-11.3 pp), discussed next.

VI. DISCUSSION AND LIMITATIONS

Why the content-KL works. The tags live at supervised positions and are predicted with the original high weights, so they are learned essentially as before. Every other position is anchored to the frozen base, which is exactly the behavior we want to preserve. The mechanism is a single-stage, per-token analogue of the “re-invoke” two-stage procedure [10] and of SDFT’s on-policy anchoring [1], but it requires neither rollouts nor a reward.

The GSM8K caveat. The distill model drops 11 pp below base on GSM8K, a long-form multi-step reasoning task. This is the honest tradeoff of a strong anchoring term ($\lambda=1.0$): content distributions are heavily pinned to the base, and on 300 samples the 4B base’s fragile chain-of-thought is sensitive to any distributional shift. Two natural follow-ups are (i) a lower λ (our sweep did not include a math benchmark), and (ii) the EMA-teacher or on-policy variants of SDFT (SPEC D3/D4), which track the student’s progress and may preserve the student’s adaptability on reasoning while still reducing forgetting.

A. Next steps: learning only part of the answer

The crux of our method is the *positional separation* of the objective: a small, well-localized set of supervised positions carries the new behavior, and every other position is anchored to the frozen base. This is a special case of

a much more common and important fine-tuning pattern—*partial-answer supervision*—where the student’s target output is already largely correct and only a small, identifiable part must be edited. Our approach should transfer directly to any such setting, and we plan to evaluate it there next. Concretely:

- **Guardrail classifiers.** The most immediate target is fine-tuning an LLM to behave as a *classifier*. For a content-moderation guardrail, the input is a text and the target output is a verdict from a small fixed taxonomy (e.g. “OK” / “FLAGGED”, or a category label), often preceded by a one-line justification. Plain SFT on this data collapses the model’s output distribution onto the label tokens and erodes general abilities. With our recipe, the *verdict position is the “tag”* (weighted cross-entropy) and the justification and everything else are the “content” (forward-KL to the frozen base). The model then learns the decision boundary while keeping its language, instruction-following and knowledge intact—exactly the retention we measured for the duplex tags (Table I).
- **Other candidate settings.** (i) *Schemafformat fix-ups* where the answer semantics are correct but the formatting (JSON keys, tool-call syntax) must change; (ii) *claim correction* (medical, legal, RAG) where only a factually wrong token span needs editing; (iii) *code repair* where only the buggy lines change; (iv) *citation insertion*.

The enabling requirement is that the “new behavior” is locatable at predictable token positions. When it is—a verdict token, a tag, a format delimiter—the off-policy recipe above is directly applicable and stays at plain-SFT cost. When the edit location is *not* known a priori, one can first align the student to a canonical template (exactly as the deterministic micro-turn builder does for duplex dialogue) so the edit collapses onto predictable positions; otherwise the on-policy SDFT variant [1] is the fallback. This planned evaluation also speaks to the principle of *RL’s Razor* [9]: forgetting is governed by the KL to the base on the new task, and partial-answer settings are precisely where an explicit per-token KL constraint is cheapest and most natural, because the supervised support is small.

Limitations. Our evaluation is on a single backbone (Qwen3-4B), a single language (pt-BR) and subset-scale benchmarks; the IFEval and GSM8K numbers carry per-sample stderr of order 2 pp. We did not ablate λ against math, nor test the full 50k-scale data regime. The offline cache assumes the teacher is fixed, which is true by construction but means the student can never “teach forward” a better protocol—acceptable here because the tags are deterministic.

VII. CONCLUSION

We showed that the duplex-cascade protocol can be learned with plain supervised fine-tuning *without* the catastrophic regression that usually accompanies it, by splitting the objective into a supervised tag term and a per-token forward-KL to the frozen base model. The approach needs neither reinforcement learning nor on-policy rollouts, and with an offline teacher cache it costs no more than SFT. On a pt-BR duplex-cascade

model it matches the plain-SFT protocol accuracy while preserving base-level function calling, general instruction following, MMLU, HellaSwag, ARC and TruthfulQA and improving IFEval—where the SFT baseline regresses broadly. Per-token KL-anchoring to the frozen base is a practical, RL-free anti-forgetting tool that generalizes beyond duplex dialogue.

REFERENCES

- [1] I. Shenfeld, M. Damani, J. Hübötter, and P. Agrawal, “On-policy self-distillation fine-tuning (SDFT): self-distillation enables continual learning,” arXiv:2601.19897, 2026.
- [2] J. Yang, Y. Fujita, and Y. Sudo, “DuplexCascade: Full-duplex speech-to-speech dialogue with VAD-free cascaded ASR–LLM–TTS pipeline and micro-turn optimization,” arXiv:2603.09180, 2026.
- [3] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proc. AISTATS*, 2011.
- [4] J. Kirkpatrick *et al.*, “Overcoming catastrophic forgetting in neural networks,” *PNAS*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [5] Z. Li and D. Hoiem, “Learning without forgetting,” *IEEE TPAMI*, vol. 40, no. 12, pp. 2935–2947, 2018.
- [6] L. Mecklenburg *et al.*, “Injecting new knowledge into large language models via supervised fine-tuning,” arXiv:2404.00213, 2024.
- [7] T. Chu, Y. Zhai, J. Yang, S. Tong, S. Xie, D. Schuurmans, Q. V. Le, S. Levine, and Y. Ma, “SFT memorizes, RL generalizes: a comparative study of foundation model post-training,” in *Proc. ICML*, 2025, pp. 10818–10838.
- [8] R. Agarwal, N. Vieillard, Y. Zhou, P. Stańczyk, S. Ramos, M. Geist, and O. Bachem, “On-policy distillation of language models: learning from self-generated mistakes,” in *Proc. ICLR*, 2024, arXiv:2306.13649.
- [9] I. Shenfeld, J. Pari, and P. Agrawal, “RL’s razor: why online reinforcement learning forgets less,” in *Proc. ICLR*, 2026, arXiv:2509.04259.
- [10] K. Lu and Thinking Machines Lab, “On-policy distillation,” Thinking Machines Lab: Connectionism, DOI 10.64434/tml.20251026, 2025.
- [11] Y. Bai *et al.*, “Constitutional AI: harmlessness from AI feedback,” arXiv:2212.08073, 2022.
- [12] C. Snell, D. Klein, and R. Zhong, “Learning by distilling context,” arXiv:2209.15189, 2022.
- [13] E. J. Hu *et al.*, “LoRA: low-rank adaptation of large language models,” in *Proc. ICLR*, 2022.
- [14] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: efficient finetuning of quantized LLMs,” in *Proc. NeurIPS*, 2023.
- [15] Unsloth AI, “Unsloth: 2x faster free finetuning,” <https://unsloth.ai>, 2026.